The texvc package*

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Abstract

This package provides all[1] LaTeX command available in MediaWiki. This includes several packages like amsmath, and adds some specific commands such as `\Reals`.

1 Provided Macros

1.1 Arrows

The first group of MediaWiki custom command (other_delimiters2) defines short hand notations for some arrows.

`\darr` Short hand notation for arrow $\downarrow$.

`\dArr` Short hand notation for arrow $\Downarrow$.

`\Darr` Short hand notation for arrow $\Downarrow$.

`\lang` Short hand notation for arrow $\langle$.

`\rang` Short hand notation for arrow $\rangle$.

`\uarr` Short hand notation for arrow $\uparrow$.

`\uArr` Short hand notation for arrow $\Uparrow$.

`\Uarr` Short hand notation for arrow $\Uparrow$.

1.2 Literals

The second group of MediaWiki custom commands (other_litereals3) defines short hand notations for some literals.

`\C` Short hand notation for literal $\mathbb{C}$. *This command is deprecated.*

`\H` Short hand notation for literal $\mathbb{H}$. *This command is deprecated.*

`\N` Short hand notation for literal $\mathbb{N}$.

`\Q` Short hand notation for literal $\mathbb{Q}$.

*This document corresponds to texvc v1.3, dated 2025/11/15.

[1] The command `\or` is only available if custom code is copied into your LaTeX-file. See page 10 for details.

`\R` Short hand notation for literal $\mathbb{R}$.

`\Z` Short hand notation for literal $\mathbb{Z}$.

`\alef` Short hand notation for literal $\aleph$.

`\alefsym` Short hand notation for literal $\aleph$.

`\Alpha` Short hand notation for literal A.

`\and` Short hand notation for literal $\wedge$. *This command is deprecated.*

`\ang` Short hand notation for literal $\angle$. *This command is deprecated.*

`\Beta` Short hand notation for literal B.

`\bull` Short hand notation for literal $\bullet$.

`\Chi` Short hand notation for literal X.

`\clubs` Short hand notation for literal $\clubsuit$.

`\cnums` Short hand notation for literal $\mathbb{C}$.

`\Complex` Short hand notation for literal $\mathbb{C}$.

`\Dagger` Short hand notation for literal $\ddagger$.

`\diamonds` Short hand notation for literal $\diamondsuit$.

`\Doteq` Short hand notation for literal $\doteqdot$.

`\doublecap` Short hand notation for literal $\Cap$.

`\doublecup` Short hand notation for literal $\Cup$.

`\empty` Short hand notation for literal $\emptyset$.

`\Epsilon` Short hand notation for literal E.

`\Eta` Short hand notation for literal H.

`\exist` Short hand notation for literal $\exists$.

`\ge` Short hand notation for literal $\geq$.

`\gggtr` Short hand notation for literal $\ggg$.

`\hAar` Short hand notation for literal $\Leftrightarrow$.

`\harr` Short hand notation for literal $\leftrightarrow$.

`\Harr` Short hand notation for literal $\Leftrightarrow$.

`\hearts` Short hand notation for literal $\heartsuit$.

`\image` Short hand notation for literal $\Im$.

`\infin` Short hand notation for literal ∞.

`\Iota` Short hand notation for literal I.

`\isin` Short hand notation for literal $\in$.

`\Kappa` Short hand notation for literal K.

`\larr` Short hand notation for literal $\leftarrow$.

`\Larr` Short hand notation for literal $\Leftarrow$.

`\lArr` Short hand notation for literal $\Leftarrow$.

`\le` Short hand notation for literal $\le$.

`\lrarr` Short hand notation for literal $\leftrightarrow$.

`\Lrarr` Short hand notation for literal $\Leftrightarrow$.

`\lrArr` Short hand notation for literal $\Leftrightarrow$.

`\Mu` Short hand notation for literal M.

`\natnums` Short hand notation for literal $\mathbb{N}$.

`\ne` Short hand notation for literal $\neq$.

`\Nu` Short hand notation for literal N.

`\O` Short hand notation for literal $\emptyset$.

`\omicron` Short hand notation for literal o.

`\Omicron` Short hand notation for literal O.

`\or` Short hand notation for literal $\vee$. *This command is deprecated.*

`\part` Short hand notation for literal ∂. *This command is deprecated.*

`\plusmn` Short hand notation for literal $\pm$.

`\rarr` Short hand notation for literal $\rightarrow$.

`\Rarr` Short hand notation for literal $\Rightarrow$.

`\rArr` Short hand notation for literal $\Rightarrow$.

`\real` Short hand notation for literal $\Re$.

`\reals` Short hand notation for literal $\mathbb{R}$.

`\Reals` Short hand notation for literal $\mathbb{R}$.

`\restriction` Short hand notation for literal $\upharpoonright$.

`\Rho` Short hand notation for literal P.

`\sdot` Short hand notation for literal $\cdot$.

`\sect` Short hand notation for literal §.

`\spades` Short hand notation for literal $\spadesuit$.

`\sub` Short hand notation for literal $\subset$.

`\sube` Short hand notation for literal $\subseteq$.

`\supe` Short hand notation for literal $\supseteq$.

`\Tau` Short hand notation for literal T.

`\thetasym` Short hand notation for literal ϑ.

`\varcoppa` Short hand notation for literal ϙ.

`\weierp` Short hand notation for literal $\wp$.

`\Zeta` Short hand notation for literal Z.

2 Deprecations

According to the decision of *Wikimedia Community User Group Math*[2] the following macros have been deprecated[3]:

1. `$`
2. `%`
3. `\and`
4. `\or`
5. `\part`
6. `\ang`
7. `\C`
8. `\H`
9. `\bold`
10. `\Bbb`

The commands 1,2,4,9,10 have never been part of this package, but were available from within Wikipedia.

[2] `https://meta.wikimedia.org/w/index.php?oldid=19705444`

[3] See `https://phabricator.wikimedia.org/T197842` for the discussion.

3 Dependencies

MediaWiki uses LaTeX macros from the following packages in addition to the standard LaTeX packages:

- amsmath
- amsfonts
- amssymb
- color
- babel
- teubner
- eurosym
- cancel
- bbold
- mhchem
- stix
- arcs

This LaTeX package requires those packages automatically, so that all commands that are usable on MediaWiki are also usable in LaTeX. For a list of the provided commands, see Section A. In addition, the MediaWiki help page for math (see `https://en.wikipedia.org/wiki/WP:Math`) provides additional details on their usage.

4 Implementation

\darr This macro does the following replacement.

```
\newcommand{\darr}{\downarrow}
```

\dArr This macro does the following replacement.

```
\newcommand{\dArr}{\Downarrow}
```

\Darr This macro does the following replacement.

```
\newcommand{\Darr}{\Downarrow}
```

\lang This macro does the following replacement.

```
\newcommand{\lang}{\langle}
```

\rang This macro does the following replacement.

```
\newcommand{\rang}{\rangle}
```

\uarr This macro does the following replacement.

```
\newcommand{\uarr}{\uparrow}
```

\uArr This macro does the following replacement.

```
\newcommand{\uArr}{\Uparrow}
```

\Uarr This macro does the following replacement.

```
\newcommand{\Uarr}{\Uparrow}
```

\C This macro does the following replacement.

```
%\newcommand{\C}{\mathbb{C}}
```

\H This macro does the following replacement.

```
\renewcommand{\H}{\mathbb{H}}
```

\N This macro does the following replacement.

```
\newcommand{\N}{\mathbb{N}}
```

\Q This macro does the following replacement.

```
\newcommand{\Q}{\mathbb{Q}}
```

\R This macro does the following replacement.

```
\newcommand{\R}{\mathbb{R}}
```

`\Z` This macro does the following replacement.

```
\newcommand{\Z}{\mathbb{Z}}
```

`\alef` This macro does the following replacement.

```
\newcommand{\alef}{\aleph}
```

`\alefsym` This macro does the following replacement.

```
\newcommand{\alefsym}{\aleph}
```

`\Alpha` This macro does the following replacement.

```
\newcommand{\Alpha}{\mathrm{A}}
```

`\and` This macro does the following replacement.

```
\renewcommand{\and}{\land}
```

`\ang` This macro does the following replacement.

```
\newcommand{\ang}{\angle}
```

`\Beta` This macro does the following replacement.

```
\newcommand{\Beta}{\mathrm{B}}
```

`\bull` This macro does the following replacement.

```
\newcommand{\bull}{\bullet}
```

`\Chi` This macro does the following replacement.

```
\newcommand{\Chi}{\mathrm{X}}
```

`\clubs` This macro does the following replacement.

```
\newcommand{\clubs}{\clubsuit}
```

`\cnums` This macro does the following replacement.

```
\newcommand{\cnums}{\mathbb{C}}
```

`\Complex` This macro does the following replacement.

```
\newcommand{\Complex}{\mathbb{C}}
```

`\Dagger` This macro does the following replacement.

```
\newcommand{\Dagger}{\ddagger}
```

\diamonds This macro does the following replacement.

```
\newcommand{\diamonds}{\diamondsuit}
```

\Doteq This macro does the following replacement.

```
\renewcommand{\Doteq}{\doteqdot}
```

\doublecap This macro does the following replacement.

```
\renewcommand{\doublecap}{\Cap}
```

\doublecup This macro does the following replacement.

```
\renewcommand{\doublecup}{\Cup}
```

\empty This macro does the following replacement.

```
\renewcommand{\empty}{\emptyset}
```

\Epsilon This macro does the following replacement.

```
\newcommand{\Epsilon}{\mathrm{E}}
```

\Eta This macro does the following replacement.

```
\newcommand{\Eta}{\mathrm{H}}
```

\exist This macro does the following replacement.

```
\newcommand{\exist}{\exists}
```

\ge This macro does the following replacement.

```
\renewcommand{\ge}{\geq}
```

\gggtr This macro does the following replacement.

```
\renewcommand{\gggtr}{\ggg}
```

\hAar This macro does the following replacement.

```
\newcommand{\hAar}{\Leftrightarrow}
```

\harr This macro does the following replacement.

```
\newcommand{\harr}{\leftrightarrow}
```

\Harr This macro does the following replacement.

```
\newcommand{\Harr}{\Leftrightarrow}
```

\hearts This macro does the following replacement.

```
\newcommand{\hearts}{\heartsuit}
```

\image This macro does the following replacement.

```
\newcommand{\image}{\Im}
```

\infin This macro does the following replacement.

```
\newcommand{\infin}{\infty}
```

\Iota This macro does the following replacement.

```
\newcommand{\Iota}{\mathrm{I}}
```

\isin This macro does the following replacement.

```
\newcommand{\isin}{\in}
```

\Kappa This macro does the following replacement.

```
\newcommand{\Kappa}{\mathrm{K}}
```

\larr This macro does the following replacement.

```
\newcommand{\larr}{\leftarrow}
```

\Larr This macro does the following replacement.

```
\newcommand{\Larr}{\Leftarrow}
```

\lArr This macro does the following replacement.

```
\newcommand{\lArr}{\Leftarrow}
```

\le This macro does the following replacement.

```
\renewcommand{\le}{\leq}
```

\lrarr This macro does the following replacement.

```
\newcommand{\lrarr}{\leftrightarrow}
```

\Lrarr This macro does the following replacement.

```
\newcommand{\Lrarr}{\Leftrightarrow}
```

\lrArr This macro does the following replacement.

```
\newcommand{\lrArr}{\Leftrightarrow}
```

\Mu This macro does the following replacement.

```
\newcommand{\Mu}{\mathrm{M}}
```

\natnums This macro does the following replacement.

```
\newcommand{\natnums}{\mathbb{N}}
```

\ne This macro does the following replacement.

```
\renewcommand{\ne}{\neq}
```

\Nu This macro does the following replacement.

```
\newcommand{\Nu}{\mathrm{N}}
```

\O This macro does the following replacement.

```
\renewcommand{\O}{\emptyset}
```

\omicron This macro does the following replacement.

```
\newcommand{\omicron}{\mathrm{o}}
```

\Omicron This macro does the following replacement.

```
\newcommand{\Omicron}{\mathrm{O}}
```

\or This is a problematic macro, since it redefines the plain TEX macro `\or`. For instance, the `\thanks` command uses a custom function to determine the footnotesymol, which relies on the availibilty of the `\or` command in math mode. Thus, the macro has to be defined after `\maketitle` was executed. However, there might be more commands that use `\or` used in mathmode. Thus we don't overwrite `\or` in this pacakge. To enable the overwriting copy the code below to an appropriate position in your LATEX-file. However, it might be easier to manually replace `\or` with `\lor` which is all what the macro above does.

```
%\let\@oldor\or
%\def\or{\ifmmode\lor\else\expandafter\@oldor\fi}
```

\part This macro does the following replacement.

```
\renewcommand{\part}{\partial}
```

\plusmn This macro does the following replacement.

```
\newcommand{\plusmn}{\pm}
```

\rarr This macro does the following replacement.

```
\newcommand{\rarr}{\rightarrow}
```

`\Rarr` This macro does the following replacement.

```
\newcommand{\Rarr}{\Rightarrow}
```

`\rArr` This macro does the following replacement.

```
\newcommand{\rArr}{\Rightarrow}
```

`\real` This macro does the following replacement.

```
\newcommand{\real}{\Re}
```

`\reals` This macro does the following replacement.

```
\newcommand{\reals}{\mathbb{R}}
```

`\Reals` This macro does the following replacement.

```
\newcommand{\Reals}{\mathbb{R}}
```

`\restriction` This macro does the following replacement.

```
\renewcommand{\restriction}{\upharpoonright}
```

`\Rho` This macro does the following replacement.

```
\newcommand{\Rho}{\mathrm{P}}
```

`\sdot` This macro does the following replacement.

```
\newcommand{\sdot}{\cdot}
```

`\sect` This macro does the following replacement.

```
\newcommand{\sect}{\S}
```

`\spades` This macro does the following replacement.

```
\newcommand{\spades}{\spadesuit}
```

`\sub` This macro does the following replacement.

```
\newcommand{\sub}{\subset}
```

`\sube` This macro does the following replacement.

```
\newcommand{\sube}{\subseteq}
```

`\supe` This macro does the following replacement.

```
\newcommand{\supe}{\supseteq}
```

`\Tau` This macro does the following replacement.

```
\newcommand{\Tau}{\mathrm{T}}
```

`\thetasym` This macro does the following replacement.

```
\newcommand{\thetasym}{\vartheta}
```

`\varcoppa` This macro does the following replacement.

```
\newcommand{\varcoppa}{\mbox{\coppa}}
```

`\weierp` This macro does the following replacement.

```
\newcommand{\weierp}{\wp}
```

`\Zeta` This macro does the following replacement.

```
\newcommand{\Zeta}{\mathrm{Z}}
```

Acknowledgements

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Change History

Index

Numbers written in italic refer to the page where the corresponding entry is described; numbers underlined refer to the code line of the definition; numbers in roman refer to the code lines where the entry is used.

[4] https://meta.wikimedia.org/wiki/Wikimedia_Community_User_Group_Math

[5] https://portal.mardi4nfdi.de/

A List of all commands supported

Section A lists all commands allowed by MediaWiki. This includes the commands provided by this package but also selected commands provided by other packages (cf. Section 3).

It was generated automatically using a script that processes a JSON file that is part of the MediaWiki Math extension.

A.1 Group `big_literals`

`\Big` is rendered as $\Big($

`\Bigg` is rendered as $\Bigg($

`\Biggl` is rendered as $\Biggl($

`\Biggr` is rendered as $\Bigg($

`\Bigl` is rendered as $\Bigl($

`\Bigr` is rendered as $\Big($

`\big` is rendered as $\big($

`\bigg` is rendered as $\bigg($

`\biggl` is rendered as $\biggl($

`\biggr` is rendered as $\bigg($

`\bigl` is rendered as $\bigl($

`\bigr` is rendered as $\big($

A.2 Group `box_functions`

`\hbox` is rendered as x

\mbox is rendered as x

\text is rendered as x

\vbox is rendered as x

A.3 Group color_function

\color is rendered as *red*

\pagecolor is not rendered.

A.4 Group declh_function

\bf is rendered as

\cal is rendered as

\it is rendered as

\rm is rendered as

A.5 Group definecolor_function

\definecolor is rendered as

A.6 Group fun_ar1

\acute is rendered as $\acute{x}$

\bar is rendered as $\bar{x}$

\bcancel is rendered as $\bcancel{x}$

\bmod is rendered as $\bmod x$

\boldsymbol is rendered as $\boldsymbol{x}$

\breve is rendered as $\breve{x}$

\cancel is rendered as $\cancel{x}$

\check is rendered as $\check{x}$

\ddot is rendered as $\ddot{x}$

\dot is rendered as $\dot{x}$

\emph is rendered as *x*

\grave is rendered as $\grave{x}$

\hat is rendered as $\hat{x}$

\hphantom is rendered as

\mathcal is rendered as

\mathclose is rendered as x

\mathfrak is rendered as $\mathfrak{x}$

`\mathit` is rendered as x

`\mathopen` is rendered as $\mathopen{x}$

`\mathord` is rendered as $\mathord{x}$

`\mathpunct` is rendered as $\mathpunct{x}$

`\mathsf` is rendered as x

`\mathtt` is rendered as $\mathtt{x}$

`\overleftarrow` is rendered as $\overleftarrow{x}$

`\overleftrightarrow` is rendered as $\overleftrightarrow{x}$

`\overline` is rendered as $\overline{x}$

`\overrightarrow` is rendered as $\overrightarrow{x}$

`\phantom` is rendered as

`\pmod` is rendered as $\pmod{x}$

`\sqrt` is rendered as $\sqrt{x}$

`\textbf` is rendered as **x**

`\textit` is rendered as *x*

`\textrm` is rendered as x

`\textsf` is rendered as x

`\texttt` is rendered as `x`

`\tilde` is rendered as $\tilde{x}$

`\underline` is rendered as $\underline{x}$

`\vec` is rendered as $\vec{x}$

`\vphantom` is rendered as

`\widehat` is rendered as $\widehat{x}$

`\widetilde` is rendered as $\widetilde{x}$

`\xcancel` is rendered as $\xcancel{x}$

A.7 Group `fun_ar1nb`

`\mathbb` is rendered as $\mathbb{x}$

`\mathbf` is rendered as $\mathbf{x}$

`\mathbin` is rendered as $\mathbin{x}$

`\mathop` is rendered as $\mathop{x}$

`\mathrel` is rendered as $\mathrel{x}$

`\mathrm` is rendered as x

`\operatorname` is rendered as x

`\overarc` is rendered as 5.0pt$\widehat{x}$

`\overbrace` is rendered as $\overbrace{x}$

`\underbrace` is rendered as $\underbrace{x}$

`\xleftarrow` is rendered as $\xleftarrow{x}$

`\xrightarrow` is rendered as $\xrightarrow{x}$

A.8 Group `fun_ar1opt`

`\sqrt` is rendered as $\sqrt{x}$

`\xleftarrow` is rendered as $\xleftarrow{x}$

`\xrightarrow` is rendered as $\xrightarrow{x}$

A.9 Group `fun_ar2`

`\binom` applied on xx is rendered as $\binom{x}{x}$

`\cancelto` applied on xx is rendered as $\cancelto{x}{x}$

`\cfrac` applied on xx is rendered as $\cfrac{x}{x}$

`\dbinom` applied on xx is rendered as $\dbinom{x}{x}$

`\dfrac` applied on xx is rendered as $\dfrac{x}{x}$

`\frac` applied on xx is rendered as $\frac{x}{x}$

`\overset` applied on xx is rendered as $\overset{x}{x}$

`\stackrel` applied on xx is rendered as $\stackrel{x}{x}$

`\tbinom` applied on xx is rendered as $\tbinom{x}{x}$

`\tfrac` applied on xx is rendered as $\tfrac{x}{x}$

`\underset` applied on xx is rendered as $\underset{x}{x}$

A.10 Group `fun_ar2nb`

`\sideset` applied on ${}^{24}_{13}\sum$ is rendered as $\sideset{^2_1}{^4_3}\sum$

A.11 Group `fun_infix`

`\atop` applied on x, y is rendered as ${x \atop y}$

`\choose` applied on x, y is rendered as ${x \choose y}$

\over applied on x, y is rendered as $\frac{x}{y}$

A.12 Group fun_mhchem

\ce is rendered as x

A.13 Group hline_function

\hline applied in a table is rendered as $\begin{array}{cc} x_{11} & x_{12} \\ \hline \end{array}$

A.14 Group latex_function_names

\Pr is rendered as $\Pr$

\arccos is rendered as $\arccos$

\arcsin is rendered as $\arcsin$

\arctan is rendered as $\arctan$

\arg is rendered as $\arg$

\cos is rendered as $\cos$

\cosh is rendered as $\cosh$

\cot is rendered as $\cot$

\coth is rendered as $\coth$

\csc is rendered as $\csc$

\deg is rendered as $\deg$

\det is rendered as $\det$

\dim is rendered as $\dim$

\exp is rendered as $\exp$

\gcd is rendered as $\gcd$

\hom is rendered as $\hom$

\inf is rendered as $\inf$

\ker is rendered as $\ker$

\lg is rendered as $\lg$

\lim is rendered as $\lim$

\liminf is rendered as $\liminf$

\limsup is rendered as $\limsup$

\ln is rendered as $\ln$

\log is rendered as $\log$

\max is rendered as $\max$

`\min` is rendered as $\min$

`\sec` is rendered as $\sec$

`\sin` is rendered as $\sin$

`\sinh` is rendered as $\sinh$

`\sup` is rendered as $\sup$

`\tan` is rendered as $\tan$

`\tanh` is rendered as $\tanh$

A.15 Group `left_function`

`\left` is rendered as $($

A.16 Group `mediawiki_function_names`

`\arccot` is rendered as $\operatorname{arccot} y$

`\arccsc` is rendered as $\operatorname{arccsc} y$

`\arcsec` is rendered as $\operatorname{arcsec} y$

`\sen` is rendered as $\operatorname{sen} y$

`\sgn` is rendered as $\operatorname{sgn} y$

A.17 Group `mhchem_bond`

`\bond` is rendered as $-$

A.18 Group `mhchem_macro_1p`

`\ce` is rendered as x

`\mathbf` is rendered as $\mathbf{x}$

A.19 Group `mhchem_macro_2p`

`\frac` applied on xx is rendered as $\frac{\mathrm{x}}{\mathrm{x}}$

`\overset` applied on xx is rendered as $\overset{\mathrm{x}}{\mathrm{x}}$

`\underset` applied on xx is rendered as $\underset{\mathrm{x}}{\mathrm{x}}$

A.20 Group `mhchem_macro_2pc`

`\color` is rendered as red

A.21 Group `mhchem_macro_2pu`

`\underbrace` is rendered as $\underbrace{x}$

A.22 Group `mhchem_single_macro`

`\Alpha` is rendered as $\Alpha$

`\Beta` is rendered as $\Beta$

`\Chi` is rendered as $\Chi$

`\Delta` is rendered as Δ

`\Epsilon` is rendered as $\Epsilon$

`\Eta` is rendered as $\Eta$

`\Gamma` is rendered as Γ

`\Iota` is rendered as $\Iota$

`\Kappa` is rendered as $\Kappa$

`\Lambda` is rendered as Λ

`\Mu` is rendered as $\Mu$

`\Nu` is rendered as $\Nu$

`\Omega` is rendered as Ω

`\Omicron` is rendered as

`\Phi` is rendered as Φ

`\Pi` is rendered as Π

`\Psi` is rendered as Ψ

`\Rho` is rendered as $\Rho$

`\Sigma` is rendered as Σ

`\Tau` is rendered as $\Tau$

`\Theta` is rendered as Θ

`\Upsilon` is rendered as Υ

`\Zeta` is rendered as $\Zeta$

`\alpha` is rendered as α

`\approx` is rendered as $\approx$

`\beta` is rendered as β

`\chi` is rendered as χ

`\circ` is rendered as $\circ$

`\delta` is rendered as δ

`\epsilon` is rendered as ε

`\eta` is rendered as η

`\gamma` is rendered as γ

`\iota` is rendered as ι

`\kappa` is rendered as κ

`\lambda` is rendered as λ

`\mu` is rendered as μ

`\nu` is rendered as ν

`\omega` is rendered as ω

`\omicron` is rendered as

`\phi` is rendered as ϕ

`\pi` is rendered as π

`\pm` is rendered as $\pm$

`\psi` is rendered as ψ

`\rho` is rendered as ρ

`\sigma` is rendered as σ

`\tau` is rendered as τ

`\theta` is rendered as ϑ

`\upsilon` is rendered as υ

`\varepsilon` is rendered as ε

`\varkappa` is rendered as $\varkappa$

`\varphi` is rendered as φ

`\varpi` is rendered as ϖ

`\varrho` is rendered as ϱ

`\varsigma` is rendered as ς

`\vartheta` is rendered as ϑ

`\zeta` is rendered as ζ

A.23 Group `nullary_macro`

`\And` is rendered as ς

`\Bbbk` is rendered as $\Bbbk$

`\Box` is rendered as $\Box$

`\Bumpeq` is rendered as $\Bumpeq$

`\Cap` is rendered as $\Cap$

`\Cup` is rendered as $\Cup$

`\Delta` is rendered as Δ

`\Diamond` is rendered as $\Diamond$

`\Finv` is rendered as $\Finv$

`\Game` is rendered as $\Game$

`\Gamma` is rendered as Γ

`\Im` is rendered as $\Im$

`\Lambda` is rendered as Λ

`\Leftarrow` is rendered as $\Leftarrow$

`\Leftrightarrow` is rendered as $\Leftrightarrow$

`\Lleftarrow` is rendered as $\Lleftarrow$

`\Longleftarrow` is rendered as $\Longleftarrow$

`\Longleftrightarrow` is rendered as $\Longleftrightarrow$

`\Longrightarrow` is rendered as $\Longrightarrow$

`\Lsh` is rendered as $\Lsh$

`\Omega` is rendered as Ω

`\P` is rendered as ¶

`\Phi` is rendered as Φ

`\Pi` is rendered as Π

`\Psi` is rendered as Ψ

`\Re` is rendered as $\Re$

`\Rightarrow` is rendered as $\Rightarrow$

`\Rrightarrow` is rendered as $\Rrightarrow$

`\Rsh` is rendered as $\Rsh$

`\S` is rendered as §

`\Sigma` is rendered as Σ

`\Subset` is rendered as $\Subset$

`\Supset` is rendered as $\Supset$

`\Theta` is rendered as Θ

`\Upsilon` is rendered as Υ

`\Vdash` is rendered as $\Vdash$

`\Vvdash` is rendered as $\Vvdash$

`\Xi` is rendered as Ξ

`\aleph` is rendered as $\aleph$

`\alpha` is rendered as α

`\amalg` is rendered as $\amalg$

`\angle` is rendered as $\angle$

`\approx` is rendered as $\approx$

`\approxeq` is rendered as $\approxeq$

`\ast` is rendered as $\ast$

`\asymp` is rendered as $\asymp$

`\backepsilon` is rendered as $\backepsilon$

`\backprime` is rendered as $\backprime$

`\backsim` is rendered as $\backsim$

`\backsimeq` is rendered as $\backsimeq$

`\barwedge` is rendered as $\barwedge$

`\because` is rendered as $\because$

`\beta` is rendered as β

`\beth` is rendered as $\beth$

`\between` is rendered as $\between$

`\bigcap` is rendered as $\bigcap$

`\bigcirc` is rendered as $\bigcirc$

`\bigcup` is rendered as $\bigcup$

`\bigodot` is rendered as $\bigodot$

`\bigoplus` is rendered as $\bigoplus$

`\bigotimes` is rendered as $\bigotimes$

`\bigsqcup` is rendered as $\bigsqcup$

`\bigstar` is rendered as $\bigstar$

`\bigtriangledown` is rendered as $\bigtriangledown$

`\bigtriangleup` is rendered as $\bigtriangleup$

`\biguplus` is rendered as $\biguplus$

`\bigvee` is rendered as $\bigvee$

`\bigwedge` is rendered as $\bigwedge$

`\blacklozenge` is rendered as $\blacklozenge$

`\blacksquare` is rendered as $\blacksquare$

`\blacktriangle` is rendered as $\blacktriangle$

`\blacktriangledown` is rendered as $\blacktriangledown$

`\blacktriangleleft` is rendered as $\blacktriangleleft$

`\blacktriangleright` is rendered as $\blacktriangleright$

`\bot` is rendered as $\bot$

`\bowtie` is rendered as $\bowtie$

`\boxdot` is rendered as $\boxdot$

`\boxminus` is rendered as $\boxminus$

`\boxplus` is rendered as $\boxplus$

`\boxtimes` is rendered as $\boxtimes$

`\bullet` is rendered as $\bullet$

`\bumpeq` is rendered as $\bumpeq$

`\cap` is rendered as $\cap$

`\cdot` is rendered as $\cdot$

`\cdots` is rendered as $\cdots$

`\centerdot` is rendered as $\centerdot$

`\checkmark` is rendered as $\checkmark$

`\chi` is rendered as χ

`\circ` is rendered as $\circ$

`\circeq` is rendered as $\circeq$

`\circlearrowleft` is rendered as $\circlearrowleft$

`\circlearrowright` is rendered as $\circlearrowright$

`\circledS` is rendered as $\circledS$

`\circledast` is rendered as $\circledast$

`\circledcirc` is rendered as $\circledcirc$

`\circleddash` is rendered as $\circleddash$

`\clubsuit` is rendered as $\clubsuit$

`\colon` is rendered as $\colon$

`\complement` is rendered as $\complement$

`\cong` is rendered as $\cong$

`\coprod` is rendered as $\coprod$

`\cup` is rendered as $\cup$

`\curlyeqprec` is rendered as $\curlyeqprec$

`\curlyeqsucc` is rendered as $\curlyeqsucc$

`\curlyvee` is rendered as $\curlyvee$

`\curlywedge` is rendered as $\curlywedge$

`\curvearrowleft` is rendered as $\curvearrowleft$

`\curvearrowright` is rendered as $\curvearrowright$

`\dagger` is rendered as $\dagger$

`\daleth` is rendered as $\daleth$

`\dashv` is rendered as $\dashv$

`\ddagger` is rendered as $\ddagger$

`\ddots` is rendered as $\ddots$

`\delta` is rendered as δ

`\diagdown` is rendered as $\diagdown$

`\diagup` is rendered as $\diagup$

`\diamond` is rendered as $\diamond$

`\diamondsuit` is rendered as $\diamondsuit$

`\digamma` is rendered as $\digamma$

`\displaystyle` is rendered as

`\div` is rendered as $\div$

`\divideontimes` is rendered as $\divideontimes$

`\doteq` is rendered as $\doteq$

`\doteqdot` is rendered as $\doteqdot$

`\dotplus` is rendered as $\dotplus$

`\dots` is rendered as $\dots$

`\dotsb` is rendered as $\dotsb$

`\dotsc` is rendered as $\dotsc$

`\dotsi` is rendered as $\dotsi$

`\dotsm` is rendered as $\dotsm$

`\dotso` is rendered as $\dotso$

`\doublebarwedge` is rendered as $\doublebarwedge$

`\downdownarrows` is rendered as $\downdownarrows$

`\downharpoonleft` is rendered as $\downharpoonleft$

`\downharpoonright` is rendered as $\downharpoonright$

`\ell` is rendered as ℓ

`\emptyset` is rendered as $\emptyset$

`\epsilon` is rendered as ϵ

`\eqcirc` is rendered as $\eqcirc$

`\eqsim` is rendered as $\eqsim$

`\eqslantgtr` is rendered as $\eqslantgtr$

`\eqslantless` is rendered as $\eqslantless$

`\equiv` is rendered as $\equiv$

`\eta` is rendered as η

`\eth` is rendered as $\eth$

`\exists` is rendered as $\exists$

`\fallingdotseq` is rendered as $\fallingdotseq$

`\flat` is rendered as $\flat$

`\forall` is rendered as $\forall$

`\frown` is rendered as $\frown$

`\gamma` is rendered as γ

`\geq` is rendered as $\geq$

`\geqq` is rendered as $\geqq$

`\geqslant` is rendered as $\geqslant$

`\gets` is rendered as $\gets$

`\gg` is rendered as $\gg$

`\ggg` is rendered as $\ggg$

`\gimel` is rendered as $\gimel$

`\gnapprox` is rendered as $\gnapprox$

`\gneq` is rendered as $\gneq$

`\gneqq` is rendered as $\gneqq$

`\gnsim` is rendered as $\gnsim$

`\gtrapprox` is rendered as $\gtrapprox$

`\gtrdot` is rendered as $\gtrdot$

`\gtreqless` is rendered as $\gtreqless$

`\gtreqqless` is rendered as $\gtreqqless$

`\gtrless` is rendered as $\gtrless$

`\gtrsim` is rendered as $\gtrsim$

`\gvertneqq` is rendered as $\gvertneqq$

`\hbar` is rendered as $\hbar$

`\heartsuit` is rendered as $\heartsuit$

`\hookleftarrow` is rendered as $\hookleftarrow$

`\hookrightarrow` is rendered as $\hookrightarrow$

`\hslash` is rendered as $\hslash$

`\iff` is rendered as $\iff$

`\iiiint` is rendered as $\iiiint$

`\iiint` is rendered as $\iiint$

`\iint` is rendered as $\iint$

`\imath` is rendered as $\imath$

`\implies` is rendered as $\implies$

`\in` is rendered as $\in$

`\infty` is rendered as ∞

`\injlim` is rendered as $\injlim$

`\int` is rendered as $\int$

`\intBar` is rendered as $\intBar$

`\intbar` is rendered as $\intbar$

`\intercal` is rendered as $\intercal$

`\iota` is rendered as ι

`\jmath` is rendered as $\jmath$

`\kappa` is rendered as κ

`\lVert` is rendered as $\lVert$

`\lambda` is rendered as λ

`\land` is rendered as $\land$

`\ldots` is rendered as $\ldots$

`\leftarrow` is rendered as $\leftarrow$

`\leftarrowtail` is rendered as $\leftarrowtail$

`\leftharpoondown` is rendered as $\leftharpoondown$

`\leftharpoonup` is rendered as $\leftharpoonup$

`\leftleftarrows` is rendered as $\leftleftarrows$

`\leftrightarrow` is rendered as $\leftrightarrow$

`\leftrightarrows` is rendered as $\leftrightarrows$

`\leftrightharpoons` is rendered as $\leftrightharpoons$

`\leftrightsquigarrow` is rendered as $\leftrightsquigarrow$

`\leftthreetimes` is rendered as $\leftthreetimes$

`\leq` is rendered as $\leq$

`\leqq` is rendered as $\leqq$

`\leqslant` is rendered as $\leqslant$

`\lessapprox` is rendered as $\lessapprox$

`\lessdot` is rendered as $\lessdot$

`\lesseqgtr` is rendered as $\lesseqgtr$

`\lesseqqgtr` is rendered as $\lesseqqgtr$

`\lessgtr` is rendered as $\lessgtr$

`\lesssim` is rendered as $\lesssim$

`\limits` is rendered for example as $\bigcap\limits_a^b$

`\ll` is rendered as $\ll$

`\lll` is rendered as $\lll$

`\lnapprox` is rendered as $\lnapprox$

`\lneq` is rendered as $\lneq$

`\lneqq` is rendered as $\lneqq$

`\lnot` is rendered as $\lnot$

`\lnsim` is rendered as $\lnsim$

`\longleftarrow` is rendered as $\longleftarrow$

`\longleftrightarrow` is rendered as $\longleftrightarrow$

`\longmapsto` is rendered as $\longmapsto$

`\longrightarrow` is rendered as $\longrightarrow$

`\looparrowleft` is rendered as $\looparrowleft$

`\looparrowright` is rendered as $\looparrowright$

`\lor` is rendered as $\lor$

`\lozenge` is rendered as $\lozenge$

`\ltimes` is rendered as $\ltimes$

`\lvertneqq` is rendered as $\lvertneqq$

`\mapsto` is rendered as $\mapsto$

`\measuredangle` is rendered as $\measuredangle$

`\mho` is rendered as $\mho$

`\mid` is rendered as $\mid$

`\mod` is rendered as $\mod$

`\models` is rendered as $\models$

`\mp` is rendered as $\mp$

`\mu` is rendered as μ

`\multimap` is rendered as $\multimap$

`\nLeftarrow` is rendered as $\nLeftarrow$

`\nLeftrightarrow` is rendered as $\nLeftrightarrow$

`\nRightarrow` is rendered as $\nRightarrow$

`\nVDash` is rendered as $\nVDash$

`\nVdash` is rendered as $\nVdash$

`\nabla` is rendered as ∇

`\natural` is rendered as $\natural$

`\ncong` is rendered as $\ncong$

`\nearrow` is rendered as $\nearrow$

`\neg` is rendered as $\neg$

`\neq` is rendered as $\neq$

`\nexists` is rendered as $\nexists$

`\ngeq` is rendered as $\ngeq$

`\ngeqq` is rendered as $\ngeqq$

`\ngeqslant` is rendered as $\ngeqslant$

`\ngtr` is rendered as $\ngtr$

`\ni` is rendered as $\ni$

`\nleftarrow` is rendered as $\nleftarrow$

`\nleftrightarrow` is rendered as $\nleftrightarrow$

`\nleq` is rendered as $\nleq$

`\nleqq` is rendered as $\nleqq$

`\nleqslant` is rendered as $\nleqslant$

`\nless` is rendered as $\nless$

`\nmid` is rendered as $\nmid$

`\nolimits` is rendered for example as $\bigcap\nolimits_a^b$

`\not` is rendered as $/$

`\notin` is rendered as $\notin$

`\nparallel` is rendered as $\nparallel$

`\nprec` is rendered as $\nprec$

`\npreceq` is rendered as $\npreceq$

`\nrightarrow` is rendered as $\nrightarrow$

`\nshortmid` is rendered as $\nshortmid$

`\nshortparallel` is rendered as $\nshortparallel$

`\nsim` is rendered as $\nsim$

`\nsubseteq` is rendered as $\nsubseteq$

`\nsubseteqq` is rendered as $\nsubseteqq$

`\nsucc` is rendered as $\nsucc$

`\nsucceq` is rendered as $\nsucceq$

`\nsupseteq` is rendered as $\nsupseteq$

`\nsupseteqq` is rendered as $\nsupseteqq$

`\ntriangleleft` is rendered as $\ntriangleleft$

`\ntrianglelefteq` is rendered as $\ntrianglelefteq$

`\ntriangleright` is rendered as $\ntriangleright$

`\ntrianglerighteq` is rendered as $\ntrianglerighteq$

`\nu` is rendered as ν

`\nvDash` is rendered as $\nvDash$

`\nvdash` is rendered as $\nvdash$

`\nwarrow` is rendered as $\nwarrow$

`\odot` is rendered as $\odot$

`\oiiint` is rendered as $\oiiint$

`\oiint` is rendered as $\oiint$

`\oint` is rendered as $\oint$

`\ointctrclockwise` is rendered as $\ointctrclockwise$

`\omega` is rendered as ω

`\ominus` is rendered as $\ominus$

`\oplus` is rendered as $\oplus$

`\oslash` is rendered as $\oslash$

`\otimes` is rendered as $\otimes$

`\parallel` is rendered as $\parallel$

`\partial` is rendered as ∂

`\perp` is rendered as $\perp$

`\phi` is rendered as ϕ

`\pi` is rendered as π

`\pitchfork` is rendered as $\pitchfork$

`\pm` is rendered as $\pm$

`\prec` is rendered as $\prec$

`\precapprox` is rendered as $\precapprox$

`\preccurlyeq` is rendered as $\preccurlyeq$

`\preceq` is rendered as $\preceq$

`\precnapprox` is rendered as $\precnapprox$

`\precneqq` is rendered as $\precneqq$

`\precnsim` is rendered as $\precnsim$

`\precsim` is rendered as $\precsim$

`\prime` is rendered as $\prime$

`\prod` is rendered as $\prod$

`\projlim` is rendered as $\projlim$

`\propto` is rendered as $\propto$

`\psi` is rendered as ψ

`\qquad` is rendered as

`\quad` is rendered as

`\rVert` is rendered as $\rVert$

`\rho` is rendered as ρ

`\rightarrow` is rendered as $\rightarrow$

`\rightarrowtail` is rendered as $\rightarrowtail$

`\rightharpoondown` is rendered as $\rightharpoondown$

`\rightharpoonup` is rendered as $\rightharpoonup$

`\rightleftarrows` is rendered as $\rightleftarrows$

`\rightrightarrows` is rendered as $\rightrightarrows$

`\rightsquigarrow` is rendered as $\rightsquigarrow$

`\rightthreetimes` is rendered as $\rightthreetimes$

`\risingdotseq` is rendered as $\risingdotseq$

`\rtimes` is rendered as $\rtimes$

`\scriptscriptstyle` is rendered as

`\scriptstyle` is rendered as

`\searrow` is rendered as $\searrow$

`\setminus` is rendered as $\setminus$

`\sharp` is rendered as $\sharp$

`\shortmid` is rendered as $\shortmid$

`\shortparallel` is rendered as $\shortparallel$

`\sigma` is rendered as σ

`\sim` is rendered as $\sim$

`\simeq` is rendered as $\simeq$

`\smallfrown` is rendered as $\smallfrown$

`\smallsetminus` is rendered as $\smallsetminus$

`\smallsmile` is rendered as $\smallsmile$

`\smile` is rendered as $\smile$

`\spadesuit` is rendered as $\spadesuit$

`\sphericalangle` is rendered as $\sphericalangle$

`\sqcap` is rendered as $\sqcap$

`\sqcup` is rendered as $\sqcup$

`\sqsubset` is rendered as $\sqsubset$

`\sqsubseteq` is rendered as $\sqsubseteq$

`\sqsupset` is rendered as $\sqsupset$

`\sqsupseteq` is rendered as $\sqsupseteq$

`\square` is rendered as $\square$

`\star` is rendered as $\star$

`\subset` is rendered as $\subset$

`\subseteq` is rendered as $\subseteq$

`\subseteqq` is rendered as $\subseteqq$

`\subsetneq` is rendered as $\subsetneq$

`\subsetneqq` is rendered as $\subsetneqq$

`\succ` is rendered as $\succ$

`\succapprox` is rendered as $\succapprox$

`\succcurlyeq` is rendered as $\succcurlyeq$

`\succeq` is rendered as $\succeq$

`\succnapprox` is rendered as $\succnapprox$

`\succneqq` is rendered as $\succneqq$

`\succnsim` is rendered as $\succnsim$

`\succsim` is rendered as $\succsim$

`\sum` is rendered as $\sum$

`\supset` is rendered as $\supset$

`\supseteq` is rendered as $\supseteq$

`\supseteqq` is rendered as $\supseteqq$

`\supsetneq` is rendered as $\supsetneq$

`\supsetneqq` is rendered as $\supsetneqq$

`\surd` is rendered as $\surd$

`\swarrow` is rendered as $\swarrow$

`\tau` is rendered as τ

`\textstyle` is rendered as

`\therefore` is rendered as $\therefore$

`\theta` is rendered as θ

`\thickapprox` is rendered as $\thickapprox$

`\thicksim` is rendered as $\thicksim$

`\times` is rendered as $\times$

`\to` is rendered as $\to$

`\top` is rendered as $\top$

`\triangle` is rendered as $\triangle$

`\triangledown` is rendered as $\triangledown$

`\triangleleft` is rendered as $\triangleleft$

`\trianglelefteq` is rendered as $\trianglelefteq$

`\triangleq` is rendered as $\triangleq$

`\triangleright` is rendered as $\triangleright$

`\trianglerighteq` is rendered as $\trianglerighteq$

`\upharpoonleft` is rendered as $\upharpoonleft$

`\upharpoonright` is rendered as $\upharpoonright$

`\uplus` is rendered as $\uplus$

`\upsilon` is rendered as υ

`\upuparrows` is rendered as $\upuparrows$

`\vDash` is rendered as $\vDash$

`\varDelta` is rendered as $\varDelta$

`\varGamma` is rendered as $\varGamma$

`\varLambda` is rendered as $\varLambda$

`\varOmega` is rendered as $\varOmega$

`\varPhi` is rendered as $\varPhi$

`\varPi` is rendered as $\varPi$

`\varSigma` is rendered as $\varSigma$

`\varTheta` is rendered as $\varTheta$

`\varUpsilon` is rendered as $\varUpsilon$

`\varXi` is rendered as $\varXi$

`\varepsilon` is rendered as ε

`\varinjlim` is rendered as $\varinjlim$

`\varkappa` is rendered as $\varkappa$

`\varliminf` is rendered as $\varliminf$

`\varlimsup` is rendered as $\varlimsup$

`\varnothing` is rendered as $\varnothing$

`\varointclockwise` is rendered as $\varointclockwise$

`\varphi` is rendered as φ

`\varpi` is rendered as ϖ

`\varprojlim` is rendered as $\varprojlim$

`\varpropto` is rendered as $\varpropto$

`\varrho` is rendered as ϱ

`\varsigma` is rendered as ς

`\varsubsetneq` is rendered as $\varsubsetneq$

`\varsubsetneqq` is rendered as $\varsubsetneqq$

`\varsupsetneq` is rendered as $\varsupsetneq$

`\varsupsetneqq` is rendered as $\varsupsetneqq$

`\vartheta` is rendered as ϑ

`\vartriangle` is rendered as $\vartriangle$

`\vartriangleleft` is rendered as $\vartriangleleft$

`\vartriangleright` is rendered as $\vartriangleright$

`\vdash` is rendered as $\vdash$

`\vdots` is rendered as $\vdots$

`\vee` is rendered as $\vee$

`\veebar` is rendered as $\veebar$

`\vline` is rendered as $|$

`\wedge` is rendered as $\wedge$

`\wp` is rendered as $\wp$

`\wr` is rendered as ≀

`\xi` is rendered as ξ

`\zeta` is rendered as ζ

A.24 Group `nullary_macro_in_mbox`

`\AA` is rendered as ∀

`\Coppa` is rendered as λ

`\Digamma` is rendered as ϵ

`\Koppa` is rendered as λ

`\Sampi` is rendered as ν

`\Stigma` is rendered as μ

`\coppa` is rendered as ι

`\euro` is rendered as *e*

`\geneuro` is rendered as €

`\geneuronarrow` is rendered as €

`\geneurowide` is rendered as €

`\koppa` is rendered as θ

`\officialeuro` is rendered as *e*

`\sampi` is rendered as σ

`\stigma` is rendered as Σ

`\textvisiblespace` is rendered as

`\varstigma` is rendered as ϒ

A.25 Group `other_delimiters1`

`\Downarrow` is rendered as $\Downarrow$

`\Uparrow` is rendered as $\Uparrow$

`\Updownarrow` is rendered as $\Updownarrow$

`\Vert` is rendered as $\Vert$

`\backslash` is rendered as $\backslash$

`\downarrow` is rendered as $\downarrow$

`\langle` is rendered as $\langle$

`\lbrace` is rendered as $\lbrace$

`\lbrack` is rendered as $\lbrack$

`\lceil` is rendered as $\lceil$

`\lfloor` is rendered as ⌊

`\llcorner` is rendered as ⌞

`\lrcorner` is rendered as ⌟

`\rangle` is rendered as ⟩

`\rbrace` is rendered as }

`\rbrack` is rendered as]

`\rceil` is rendered as ⌉

`\rfloor` is rendered as ⌋

`\rightleftharpoons` is rendered as ⇌

`\twoheadleftarrow` is rendered as ↞

`\twoheadrightarrow` is rendered as ↠

`\ulcorner` is rendered as ⌜

`\uparrow` is rendered as ↑

`\updownarrow` is rendered as ↕

`\urcorner` is rendered as ⌝

`\vert` is rendered as |

A.26 Group `other_delimiters2`

`\Darr` is rendered as ⇓

`\Uarr` is rendered as ⇑

`\dArr` is rendered as ⇓

`\darr` is rendered as ↓

`\lang` is rendered as ⟨

`\rang` is rendered as ⟩

`\uArr` is rendered as ⇑

`\uarr` is rendered as ↑

A.27 Group `right_function`

`\right` is rendered as)